Worksheet No. - 6

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Aim/Overview of the practical:

Learn how to create, query, and manage views in SQL to simplify database queries and provide a layer of abstraction for end-users.

Objective:

- **Data Abstraction:** To understand how to hide complex table joins and calculations behind a simple virtual table interface.
- **Enhanced Security:** To learn how to restrict user access to sensitive columns by providing views instead of direct table access.
- **Query Simplification:** To master the creation of views that pre-join multiple tables, making reporting easier for non-technical users.
- **View Management:** To understand the syntax for creating, altering, and dropping views, as well as the naming conventions required for efficient data access.

Theory:

A View is essentially a virtual table based on the result-set of an SQL statement. It does not contain data of its own but dynamically pulls data from the underlying "base tables".

1. **Simple Views:** Created from a single table without any aggregate functions or grouping. These are often updatable.
2. **Complex Views:** Created from multiple tables using JOINs, or including GROUP BY and aggregate functions. These provide a consolidated summary of the database.
3. **Security Layer:** In enterprise environments, views are used to grant permissions on specific subsets of data. For example, a "SalaryView" might exclude the "Employee_SSN" or "Home_Address" columns for privacy.
4. **Benefits:** They simplify the user experience, ensure data consistency across reports, and reduce the risk of accidental data modification by providing read-only abstractions.

Input/Apparatus Used:

- Oracle
- MS SSMS
- PostgreSQL

Procedure/Algorithm/Code :

```
-- Create table
CREATE TABLE Employee (
   id INT PRIMARY KEY,
   name VARCHAR(100),
   department VARCHAR(50),
   managerId INT
);

-- Insert data
INSERT INTO Employee (id, name, department, managerId) VALUES
(101, 'John', 'A', NULL),
(102, 'Dan', 'A', 101),
(103, 'James', 'A', 101),
(104, 'Amy', 'A', 101),
(105, 'Anne', 'A', 101),
(106, 'Ron', 'B', 101);


SELECT e1.name
FROM Employee e1
```

```
JOIN Employee e2
  ON e1.id = e2.managerId
GROUP BY e1.id, e1.name
HAVING COUNT(e2.id) >= 5;
```

Output:

	name character varying (100)
1	John

Learning outcomes (What I have learnt):

- **Abstraction Proficiency:** Students will be able to create and query views to simplify efficient data access and abstraction.
- **Security Implementation:** Students will understand how to use views for data masking and providing restricted access to sensitive information.
- **Syntactic Accuracy:** Students will demonstrate the correct syntax for creating and querying views, ensuring logical clarity in naming conventions.
- **Real-world Application:** Students will be able to design views for practical domains like Library Management Systems or Payroll Systems to demonstrate functionality.